

PocketSmart AI: AI-Powered Budget & Recommendation Assistant

PocketSmart AI | Team ID: 6ab222a5e880316a482c6374 | Date: 30 September 2026

Date	30 September 2026
Team ID	6ab222a5e880316a482c6374
Project Name	PocketSmart AI
Team Members	Joison V, Gurun J, Sanjai Y, Aswarth U, Yaswanth N
Maximum Marks	Documentation

Project Description

PocketSmart AI is an AI-powered web application designed to help users plan everyday lifestyle needs within a selected budget. It provides personalized recommendations for home interiors, party planning and jewelry selection. Users enter budget, preferences, occasion and, for jewelry, an optional outfit image. FastAPI handles the application flow while Gemini AI generates relevant recommendations.

The application also connects its recommendation concept with popular shopping and service platforms such as Amazon, Flipkart, IKEA, Swiggy, Zomato and OYO, helping users reduce the time and effort required to compare options.

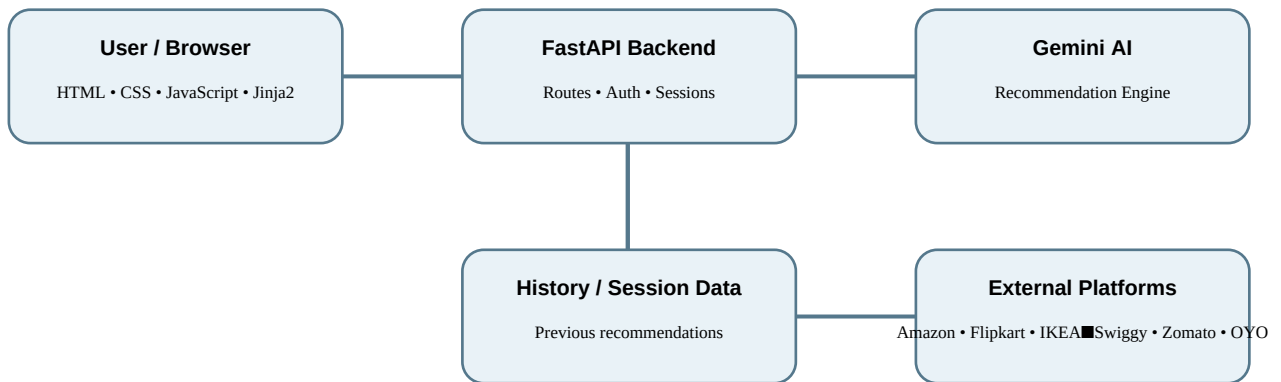
Scenarios

Scenario 1: A user wants to improve a room within a fixed budget. The user selects Home Planner, enters room details and quantities, and receives suggestions for furniture, lighting and decor.

Scenario 2: A user is planning a party and needs to control spending. The user enters the event details, guest count and budget, and receives food, venue and decoration suggestions.

Scenario 3: A user wants jewelry for an occasion. The user selects a style and budget and may upload an outfit image so the AI can use the image context when generating recommendations.

Technical Architecture



Description: PocketSmart AI uses a layered architecture in which the frontend collects input, FastAPI validates and routes requests, Gemini AI interprets the context and generates recommendations, and the frontend displays the results. Authentication/session handling and history provide continuity for users.

Pre-requisites

- Python programming knowledge.
- FastAPI and Uvicorn setup.
- HTML, CSS and JavaScript fundamentals.
- Jinja2 template knowledge.
- Gemini AI/API configuration.
- Basic Git and environment-variable management.

Project Workflow

Milestone 1: Problem Understanding and Architecture

- Identify budget-planning and recommendation pain points.
- Define Home, Party and Jewelry planner modules.
- Select FastAPI, Gemini AI and the web frontend stack.
- Define the layered architecture and data flow.

Milestone 2: Core Functionality Development

- Implement registration, login and logout.
- Develop Home Planner.
- Develop Party Planner.
- Develop Jewelry Planner with optional image input.
- Connect planner requests to the Gemini recommendation layer.

Milestone 3: Backend and AI Integration

- Create separate API routes for each planner.
- Validate incoming user data.
- Build structured AI context from budget and preferences.
- Return structured recommendations to the frontend.

Milestone 4: Frontend Development

- Create planner forms and result views.
- Use Jinja2 templates with HTML/CSS/JavaScript.
- Display recommendations clearly and consistently.
- Provide history access.

Milestone 5: Testing and Documentation

- Test different budgets and room/product quantities.
- Test party types and guest counts.
- Test jewelry with and without an image.
- Test authentication and history.
- Document limitations and future improvements.

Exploring the Website's Web Pages

Home Page / Dashboard



Figure 1: PocketSmart AI Home Page

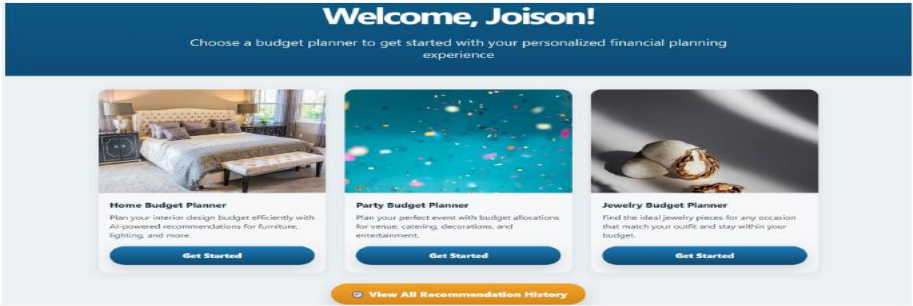


Figure 2: User Dashboard

Figure: Home page and dashboard screenshots from the supplied PocketSmart AI documentation.

Planner Pages

Home Interior Budget Planner

Create a customized budget plan for your dream home interior

Budget Details

Total Budget (\$)

Enter your budget

Fixtures & Furniture

Number of Lights/Fixtures

Number of Lights

Number of Furniture Pieces

Furniture pieces

Number of Ceiling Fans

Number of fans

Number of Dining Tables

Dining tables

Rooms to Include

Living Room

Kitchen

Bedroom

Additional Information

Figure 3: Home Interior Budget Planner

Party Budget Planner

Plan your perfect event with AI-powered budget recommendations

Basic Information

Total Budget (\$)

Enter your budget

Number of Guests

Number of guests

Event Details

Party Type

Birthday, wedding, etc.

Venue Type

Home, hall, outdoor, etc.

Party Needs

Catering

Decoration

Entertainment

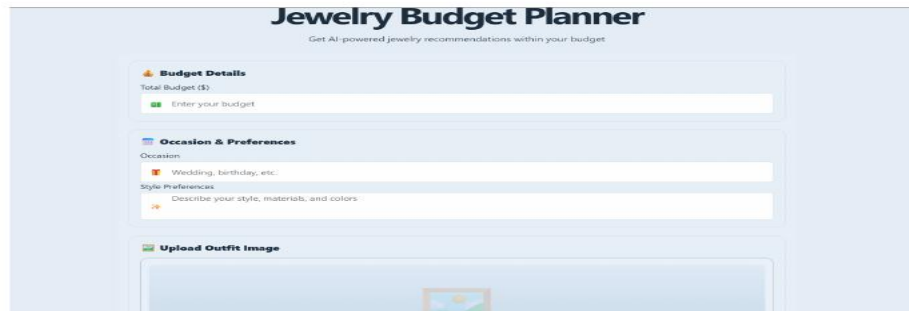
Additional Requirements

Special requests, themes, dietary

Figure 4: Party Budget Planner

Figure: Home Interior and Party Budget Planner screenshots.

Jewelry Planner and History



Jewelry Budget Planner
Get AI-powered jewelry recommendations within your budget

Budget Details
Total Budget (\$):

Occasion & Preferences
Occasion:

Style Preferences:

Upload Outfit Image

Figure 5: Jewelry Budget Planner



Upload Outfit Image

Get Recommendations

Figure 5 (continued): Jewelry Budget Planner – Recommendations



Recent Activity	
	Home Budget Plan Budget: \$5,000 - Created on May 7, 2025
	Birthday Party Budget Plan Budget: \$1,200 for 30 guests - Created on May 5, 2025
	Jewelry Budget Plan Budget: \$800 for Anniversary - Created on April 28, 2025

Figure 6: Recommendation History

Figure: Jewelry Planner, recommendations and history screenshots.

Important API Routes

Route	Purpose
/generate-home	Generate home interior recommendations
/generate-party	Generate party planning recommendations
/generate-jewelry	Generate jewelry recommendations
/register	Create user account
/login	Authenticate existing user
/logout	End session
/history	Retrieve previous recommendation activity

Testing

- Different budget values were considered across planners.
- Home planner testing includes room and product quantity variations.
- Party planner testing includes party types and guest counts.
- Jewelry planner testing includes flows with and without an image.
- Authentication and recommendation history are included in testing.

Advantages

- Saves planning time.
- Provides budget-aware suggestions.
- Supports multiple lifestyle categories in one application.
- Supports optional image input for jewelry recommendations.
- Provides a personalized experience through sessions and history.

Limitations

- Recommendation quality depends on AI responses and available product information.
- External platform information can change over time.
- Internet connectivity and valid AI/API configuration are required.
- AI-generated recommendations should be checked before real purchases.

Conclusion

PocketSmart AI demonstrates how generative AI and web technologies can be combined to simplify everyday budget planning. The application brings home interiors, party planning and jewelry selection into a single interface, while FastAPI manages the backend and Gemini AI generates personalized recommendations. The documented architecture, modules, testing approach and future enhancements provide a foundation for extending the system with more categories, languages, voice interaction, stronger price comparison and real-time availability tracking.

Team Members

S.No	Team Member
1	Joison V
2	Gurun J
3	Sanjai Y
4	Aswarth U
5	Yaswanth N